

The 1970s also witnessed further regulation of hams, with hams having to take exams to obtain licences before they were allowed to transmit. This was done first in the US, but other countries quickly followed.

By the 1980s, hams had started integrating computers into their systems. Advanced techniques such as digital signal processing were applied by hams to amateur radio services, pioneering many developments in the world of wireless communication such as phase shift keying (a modulation technique used for digital audio signals).

In India, amateur radio started in the 1920s, and by 1930 there were 16 licensed hams. Hams played a crucial role during the Indian independence movement, as various amateur radio broadcasts were set up illegally against the government. Notable among these is the Azad Hind radio set up by Indian ham Nariman Abarbad Printer. He broadcasted Gandhian protest music and uncensored news, for which he was immediately arrested and his equipment seized. The Congress utilised amateur radio services to coordinate activities with the grass root level workers during the Quit India movement. Hams also provided valuable communication support for the resistance and the invading Indian National Army under Subhash Chandra Bose.

Post independence, India's first organisation of hams, the Amateur Radio Club of India, was set up in 1948. It was soon converted to the Amateur Radio Society of India and is today the primary amateur radio organisation in India. By 1980, there were 1,600 licensed hams in India. With the waiving of import

duty on radio equipment in 1984, the number of hams started rising steadily. Hams continued to play a major role in most emergencies. In the 1991 Gulf War, a lone Indian ham in



An amateur radio operator in India